2 A drag car is a racing car that is powered by a rocket engine.

A drag car accelerates uniformly from rest until it reaches the finishing line. The engine is then switched off and a parachute opens. The car decelerates until it stops.

Fig. 2.1 shows a drag car decelerating after a race.

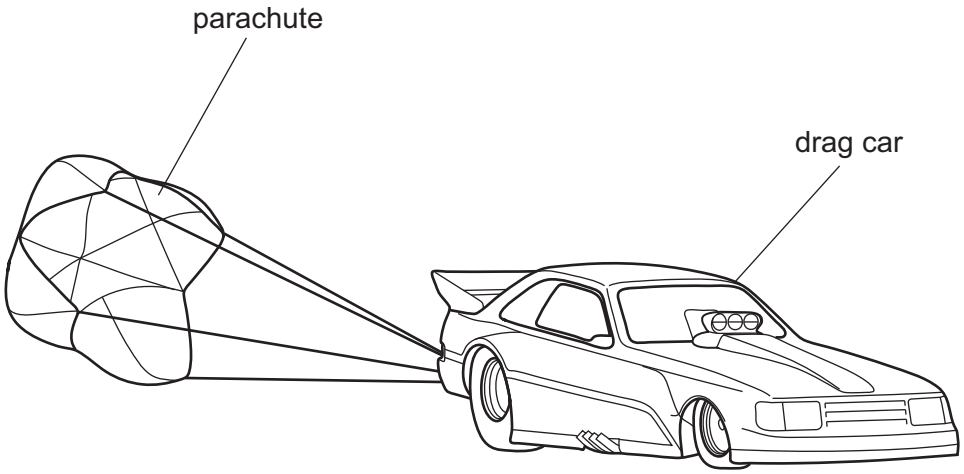

Fig. 2.1

This drag car has a mass of 1400 kg.

Fig. 2.2 is the speed–time graph for the car during a race on a straight horizontal track.

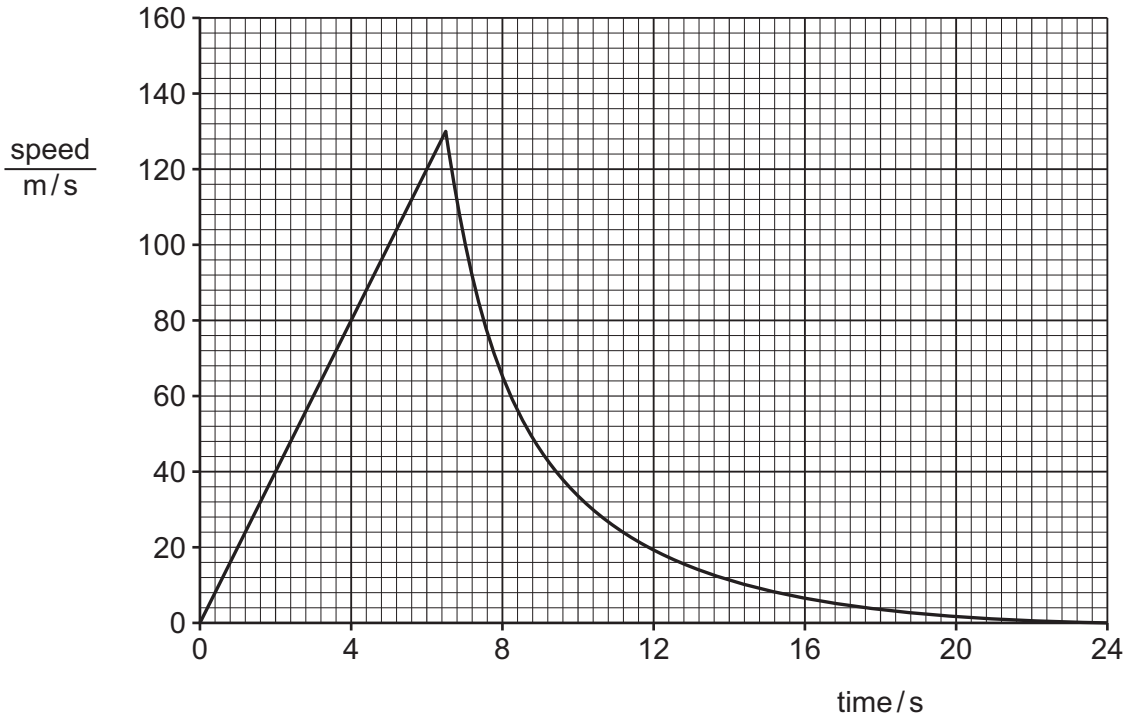

Fig. 2.2

The car reaches its maximum speed of 130 m / s at a time of 6.5 s.

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(a) **(i)** Calculate the maximum momentum of the car during the race.

maximum momentum = ... [2]

(ii) State the feature of Fig. 2.2 that represents the distance travelled by the car.

.....

..... [1]

(iii) Determine the distance travelled by the car in the first 6.5 s.

distance = ... [2]

(b) The parachute opens at 6.5 s and the car decelerates.

Describe how Fig. 2.2 shows that, after 6.5 s:

(i) the car decelerates

.....

..... [1]

(ii) the deceleration of the car is **not** constant.

.....

..... [1]

(c) Describe the energy transfer that takes place as the car slows down.

.....

..... [2]

[Total: 9]

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